

Syllabus

Spring semester 2026

Course Title	General Astronomy Lab		Course Code		PE3202		Credit	2	Classroom	Room 5903, Changjo Bldg.
			Type		Lab 2	Hours	2			
Teacher	Name	Class no.	Office	Office Hour*	Contact		Google Classroom Code			
	In-Ok Song	1,2	5306	Mon3,4 Wed2,3	051 606 2332		(Tue) gtfpbxlt (Fri) 4ruczmtn			

*Office Hour: Time for students to ask questions and discuss lecture or lab topics with teachers for deeper understanding.

1. Course Descriptions

This class goes with that of General Astronomy and will help to understand the universe – the sky, stars, galaxies, and cosmos. Each laboratory activity is required for about 2 hours per week. You will be in the classroom most of the time; however, you are expected to go out to the natural laboratory – the sky. The report will be submitted after the lab. When you use the astronomical observatory in KSA, you should follow the given instruction.

1. To introduce the universe and scientific methods of astronomy – observation, theory, and prediction.
2. To provide you opportunities to develop self-motivated study and collaborative study

2. Core Target Competencies

Subject Core Competencies				
Analytical and Creative Thinking	Leadership and Followership	Digital/ Quantitative Literacy	Adaptability/ Progression/ Spirit of Adventure	Habits/ Mindset
Bringing these elements together, the essential competencies for today's world include the ability to think both analytically and creatively in order to understand complex problems and generate innovative solutions. Equally important is the capacity to exercise leadership when needed and to practice effective followership in collaborative settings. Strong digital and quantitative literacy enables individuals to interpret, analyze, and handle data responsibly and strategically. In rapidly changing environments, adaptability, continuous progression, and a spirit of adventure empower individuals to embrace challenges and pursue growth. Ultimately, these competencies are sustained by productive habits and a reflective, open mindset that supports lifelong learning and meaningful contribution to society.				

3. Textbooks and References

A. Textbook: Edited materials

B. References

- Horizons: Exploring the universe, 14/E, M.A. Seeds and D.E. Backman (Brooks/Cole, 2017)
- Contemporary Activities in ASTRONOMY, Darrel B. Hoff & Jeffrey A. Wilkerson, Kendall Hunt
- INTRODUCTORY ASTRONOMY& ASTROPHYSICS, ZEILIK• GREGORY, Thomson Learning
- 천문학 실습, 김철희, 시그마프레스

4. Class Method

- A. Half of the experiments are composed of activities viewing the sky with the naked eye, a telescope, and a DSLR camera. Students learn to observe the Sun in the daytime and deep sky objects at night.
- B. Other experiments are about measuring distances in the universe. From nearby stars to distant galaxies, students learn to determine the distances of various celestial bodies' data step by step, so that they can get an idea of the size of the universe.
- C. Reports are in two formats, worksheet format, and full report format. As a case, follow the attached rubrics.
- ※ **For class-related information, please refer to Google Classroom. For example, announcements or report submissions, etc..**

5. Evaluation

A. Evaluation Criteria

Reports	Attitude	Attendance	Sum
80	10	10	100

B. Evaluation Method

1) Report 80%:

- In time you will get the score range of A to D, and it will be normally 1 week and the exact deadline will be announced in the classroom.
- If you submit your report after, then and within 1 weeks then you will get a score ranging from B to F. After 1 weeks, any report cannot be accepted and the score is automatically F.
- If you have any problems, please consult with the instructor in advance. A Rubric is attached.
- **Plagiarism**; a meeting of the faculty committee will be open for the FAIL of the course.

(3) Attitude 10%

- Experiment progresses: Laboratory safety instructions.
- Cooperation: in team activity, sincerity, honesty, fairness, and respect.

(4) Attendance 10%

- Unexcused absence makes the score degraded by 20% each time.
- If 4 times of unexcused absence, a meeting of the faculty committee will be open for FAIL of the course.

(5) Level of Evaluation

More than expected	A	100 - 85
Expected	B	84 - 70
Reluctantly Followed	C/D	69 - 40
Fail	F	39 - 0

(6) Criteria for awarding recognized scores to absent students

- When calculable, converting the Z-score from the attended examination to the equivalent Z-score of the missed examination. (응시고사의 표준점수 Z를 미응시고사 표준점수로 환산)
- When not calculable, In accordance with Article 10, Paragraphs 2 and 3 of the Academic Performance Management Regulations. (학업성적관리규정 제 10조 2,3항을 준용함)

6. Weekly Plans

Week (period)	Topic and Contents	(Notes)
Week 1 2.23.~3.1.	Introduction	
Core Competencies		
Week 2 3.2.~3.8.	Observation of Constellation with the naked eyes	- Seasonal Constellations - Polaris - Coordinates; Horizontal & Equatorial coordinates
Core Competencies	Leadership and Followership; Adaptability; Spirit of Adventure	
Week 3 3.9.~3.15.	Observation of Changes in Planet Locations	- Motion of planets - Planets we can observe
Core Competencies	Leadership and Followership; Adaptability; Spirit of Adventure	
Week 4 3.16.~3.22.	Observation of Changes in Planet Locations	- Motion of planets - Planets we can observe
Core Competencies	Leadership and Followership; Adaptability; Spirit of Adventure	
Week 5 3.23.~3.29.	Handy Telescope	- Telescope; refraction and reflection
Core Competencies	Analytical and Creative Thinking; Spirit of Adventure; Habits	
Week 6 3.30.~4.5.	Sunspots	- Telescopes; Mounts - Sunspot - How to observe the Sun; ND Filter, Projection, Ha filter
Core Competencies	Analytical and Creative Thinking; Leadership and Followership; Digital/ Quantitative Literacy; Adaptability; Spirit of Adventure	
Week 7 4.6.~4.12.	DSLR Astrophotography	- DSLR camera - Astrophotography - List of celestial objects observable today
Core Competencies	Leadership and Followership; Digital/ Quantitative Literacy; Adaptability; Spirit of Adventure; Habits	
Week 8 4.13.~4.19.	Midterm Exams (4.14.~4.17.)	4.13. Regular Classes

Week 9 4.20.~4.26.	Astrophotography – Startrail	- Fixed shooting
Core Competencies	Leadership and Followership; Digital/ Quantitative Literacy; Adaptability; Spirit of Adventure; Habits	
Week 10 4.27.~5.3.	Analysis of Solar spectra - Fraunhofer Lines	- Kirchhof's laws - Fraunhofer Lines
Core Competencies	Analytical and Creative Thinking; Digital/ Quantitative Literacy	
Week 11 5.4.~5.10.	Distance Determination – P-L relation of variable stars	5.5. Children's Day
Core Competencies	Analytical and Creative Thinking; Digital/ Quantitative Literacy	
Week 12 5.11.~5.17.	Distance Determination - Main Sequence Fitting	- HRD of star clusters - Distance Modulus
Core Competencies	Analytical and Creative Thinking; Digital/ Quantitative Literacy	
Week 13 5.18.~5.24.	Distance Determination - Super Nova	-Type Ia supernova
Core Competencies	Analytical and Creative Thinking; Digital/ Quantitative Literacy	
Week 14 5.25.~5.31.	Distance Determination - Pulsars	5.25.
Core Competencies	Analytical and Creative Thinking; Digital/ Quantitative Literacy	Substitute Holiday
Week 15 6.1.~6.7.	Distance Determination – Hubble-Lemaître's Law	- Hubble-Lemaître's Law
Core Competencies	Analytical and Creative Thinking; Digital/ Quantitative Literacy	
Week 16 6.8.~6.14.	Make-up class week	
Week 17 6.15.~6.19.	Final Exams (6.18.~6.19.)	

※ The above plans may change according to the academic schedule.

How to write an effective report

Almost **IMRAD** format:

Title and your name - The title should reflect the factual content of the paper. Here you will write your name and the date on which the experiment was performed or the date the lab report was written.

Filename – “RPT##_XX-XXX_yourname_title.xxx”

Introduction or Purpose – It should provide the reader with information necessary to understand the scientific basis of the experiment and the techniques used. In most cases, you should include background information and explain the theory behind the techniques. The most common mistake that students make is references. When you refer any resources from others, you should not the source of the information. You are not supposed to write introduction in the amount of half of the report. Keep in mind to write concisely key information and key technique.

Material and Methods –It should be included all raw data, observations, calculations and appropriate graphs.

Results and Analysis – A summary of the final data or output. A short paragraph should be sufficient, along with any relevant charts and graphs labeled well. Remember to title and provide legends for all graphs and tables. The graphs and tables should be comprehensible independently of their association with the text.

Summary or Discussion - Discuss the experiment and the results obtained. This does not mean you simply report the results again, but rather interpret and discuss their significance. If problems were encountered during the course of the experiment, how might they be rectified in the future? Are there any other things we could do to make this a better experiment or to more specifically address the initial question posed? Are there any better techniques available that would allow one to more accurately generate data? Is there more than one way to explain the results?

Reference Citations - As required in all scientific literature, statements of fact, not considered "common" knowledge, must be properly referenced.

- Note 1: This kind of IMRAD format is a guideline which helps you to write a report. You don't have to follow this scheme, but you can rather re-organize the report depending on the topic if you want.
- Note 2: This guideline is adapted from one in the Writing Center, the University of Wisconsin and modified for this class suitably

Song's rule of thumb:
What I did...
Because...
The methods I used are...
The results are...
Which means....

Rubric for Report

Student' Name:

Credit:

1. Organization
 - A. Directly relevant
 - B. Somewhat relevant
 - C. Remotely related/Totally unrelated
2. Quality of Information
 - A. Good organization, events are logically ordered, sharp sense of beginning and end
 - B. Organized, events are somewhat jumpy
 - C. Some organization, events jump around, start and end are unclear
3. Interest Level
 - A. Vocabulary varied, supporting details vivid
 - B. Vocabulary is varied, supporting details need work
 - C. Vocabulary is constant, Needs descriptive words
4. Neatness
 - A. Figures and tables have a caption if there is; References are given if you cited any work from others
 - B. No caption; References are given if you cited any work from others
 - C. No references are given if you cited any work from others

NOTE:

- Report header should contain
 - Student ID and **Name:**
 - Topic of the Report:
 - Submitted Date:
- **Rubric for Report** is adapted from one in San Diego County and modified for this class suitably.

Collaboration Rubric

A - Thorough Understanding

- Consistently and actively works toward group goals.
- Is sensitive to the feelings and learning needs of all group members.
- Willingly accepts and fulfills individual role within the group.
- Consistently and actively contributes knowledge, opinions, and skills.
- Values the knowledge, opinion and skills of all group members and encourages their contribution.
- Helps group identify necessary changes and encourages group action for change.

A - Good Understanding

- Works toward group goals without prompting.
- Accepts and fulfills individual role within the group.
- Contributes knowledge, opinions, and skills without prompting.
- Shows sensitivity to the feelings of others.
- Willingly participates in needed changes.

B - Satisfactory Understanding

- Works toward group goals with occasional prompting.
- Contributes to the group with occasional prompting.
- Shows sensitivity to the feelings of others.
- Participates in needed changes, with occasional prompting.

C - Needs Improvement

- Works toward group goals only when prompted.
- Contributes to the group only when prompted.
- Needs occasional reminders to be sensitive to the feelings of others.
- Participates in needed changes when prompted and encouraged.

Collaboration rubric is adapted from <Teacher Planet> community

천문대 사용 규칙

1. 관측

천문대 안전규칙에 따라 행동한다.

2. 관측후 정리

A. 모든 마운트 전원 꺼짐 확인

이유: 전원 켜 있을 경우 별이 진 후에도 별을 추적하고자 마운트 기어 시스템에 무리가 감

B. 관측일지 작성

천문대 안전 규칙

1. 천문대에서는 항상 사용자 자신이 “책임” 이 있다고 생각하고 관측을 수행한다. (이 천문대는 선배들이 활발히 사용하였으며 아꼈고, 후배들에게도 그런 천문대를 물려주길 바라는 마음이다.)
2. 천문대에 들어왔을 때, 교원 (혹은 조교)으로부터 지시를 받은 후에 사용 가능하다. 그 이전에는 절대 아무것도 만지지 않는다.
3. 망원경을 고정하고 있는 마운트 혹은 삼각대, 망원경의 방향/위치를 잘 파악하여 부딪히거나 넘어지지 않도록 주의한다.
4. 망원경이 있는 영역 (슬라이딩 돔 영역과 원형 돔 영역)에서는 음식을 먹거나, 음료를 마시거나, 야단법석을 떨지 않는다. (어두워서 물체의 위치 파악이 어려우므로 쏟거나 다른 친구에게 피해가 될 수 있으므로 이것은 피하도록 한다.)
5. 만약 안전하지 않은 상황이 발견되면 즉시 담당교원 혹은 조교선생님께 알린다.
6. 슬리퍼 착용은 천문대 입실을 금지한다. 예외 없음.
7. 낮에는 **절대** 망원경으로 태양을 관측하지 않는다. 태양 필터를 장착한 후에만 관측 가능하다